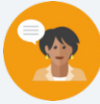


9.9 Surface Area in the Real World

Lesson Introduction

 Benchmark	MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.	 Learning Targets	- I can solve real-world problems involving the surface areas of cylinders, pyramids, prisms, cones, and spheres. - I can solve real-world problems involving the surface areas of composite figures.	
 Benchmark Coherence	Building On		Working Toward	
	MA.7.GR.2.1 Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure’s net.		N/A	
 Essential Questions	- How can I solve real-world problems involving the surface areas of cylinders, pyramids, prisms, cones, and spheres? - How can I solve real-world problems involving the surface areas of composite figures? - How is surface area used in the real-world?			
 Lesson Narrative	Lesson 9 builds on Lesson 8 in working with the surface areas of various three-dimensional figures. The purpose of Lesson 9.9 is to specifically work with real-world problems and scenarios and to be able to apply the surface area formulas in a real-world context. This lesson will guide learning in Lesson 11 on dimension changes.			
 Component Summary	9.9.1 Warm-Up (~5 min.)	Analyzing a math magic trick (<i>SE p. 100</i>)	9.9.3 Guided Instruction (10-15 min.)	Solving problems on frosting cakes using surface area (<i>SE p. 102</i>)
	9.9.2 Collaboration (10-15 min.)	Solving a packaging problem using surface area (<i>SE p. 101</i>)	9.9.4 Guided Practice (5-10 min.)	Using surface area to solve problems (<i>SE p. 103</i>)
	9.9.5 Wrap-Up (~5 min.)	Ticket Out the Door on real-world surface area (<i>SE p. 104</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Colored Pens/Pencils	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- On 9.9.2 Collaboration question 3, students may forget to divide by 100 to find the cost of the packaging. - On 9.9.3 Guided Instruction question 2, students may try to find the surface area of the rectangular prism instead of lateral area. - Students may multiply by $\frac{1}{3}$ instead of dividing on 9.9.4 Guided Practice, question 1.	Consider letting students work with either partners or groups on the whole lesson, with the exceptions of the 9.9.1 Warm-Up and 9.9.5 Wrap-Up, as these questions are more complex than the previous lesson’s questions.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Number Talk Consider using the 9.9.1 Warm-Up as an opportunity to discuss the process of how to find the “magic number.” Have students rely on the structure and the algorithm and discuss why the ending result is always 5.	MA.K12.MTR.6.1: Reasonableness Many of the problems in this lesson are real-world scenarios and require students to take out parts of the formulas to find the correct answers. Have students discuss and work to determine whether a solution is reasonable for a given context.
 Differentiate Instruction	Consider making the 9.9.2 Collaboration, 9.9.3 Guided Instruction, and the 9.9.4 Guided Practice three separate groups. Assign each student to one of the three sections. Give students a set amount of time to complete the component. Then, create new groups of three made up of one student from each of the three sections above (9.9.2 Collaboration, 9.9.3 Guided Instruction, and 9.9.4 Guided Practice). Students share their results with their group and become “experts” on their component.	
Lesson Notes:		

9.9.1 Warm-Up: Math Magic

Component Overview: Students complete a “magic math trick” and then try to justify the “trick” by creating an equation.



9.9.1 Warm-Up: Math Magic

Arthur Benjamin (b. 1961) is a professor of mathematics at Harvey Mudd College in California. Dr. Benjamin is often referred to as a “mathemagician” because he is known for his quick mental computational skills. Dr. Benjamin’s TED talks have been viewed millions of times on the internet.

Try one of Dr. Benjamin’s math “magic tricks.”

- Pick a number between 1 and 10.
- Double your number.
- Add 10 to your number.
- Divide by 2.
- Subtract the number you started with originally.

1. What number did you get?

5

2. Compare your number with your partner’s number.
What do you notice?

The answers are the same.

3. Why do you think the “trick” works?

Answers will vary. Sample response: Regardless of the first number, the process is the same.

<i>Pick a number between 1 and 10.</i>	x
<i>Double your number.</i>	$2x$
<i>Add 10 to your number.</i>	$2x + 10$
<i>Divide by 2.</i>	$x + 5$
<i>Subtract the number you started with originally.</i>	5



If time permits, have students try the steps using a number greater than 10 to see if the result is still 5.

9.9.2 Collaboration: The Best Package

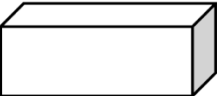
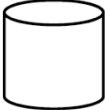
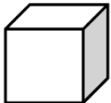
Component Overview: Students work through a scenario where three types of packaging are being used. Students will determine the surface areas of the types of packaging and then determine which type of packaging will be the cheapest. Allow students to work in partners or groups to complete this component.



9.9.2 Collaboration: The Best Package

Manuel is a graphic designer and works with various types of packaging. For the artwork, Manuel needs a container with a surface area of at least 250 square inches (in.) to ship the company's products.

- The three types of packaging that the company has in stock are shown in the table. Find the surface area for each type of packaging.

Packaging	Dimensions	Surface Area
	l : 7 in. w : 10 in. h : 4 in.	$SA = 2B + Ph$ $SA = 276 \text{ in}^2$
	d : 8 in. h : 8 in.	$SA = 2B + Ch$ $SA = 301.593 \text{ in}^2$
	s : 12 in.	$SA = 6s^2$ $SA = 864 \text{ in}^2$

- Rank the packaging's surface area from smallest to largest.

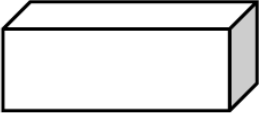
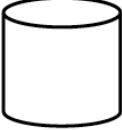
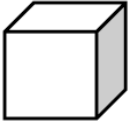
Rank	Packaging
1 (smallest)	<i>Rectangular Prism</i>
2	<i>Cylinder</i>
3 (largest)	<i>Cube</i>



Have students make a prediction to see which type of packaging they think will be the largest and smallest.

9.9.2 Collaboration: The Best Package (continued)

3. The prices for packaging are listed in the table. Based on the surface area of each package, find the cost of each type of packaging.

Packaging	Pricing	Cost of Packaging
Rectangular Prism 	\$1.75 per 100 square inches	$\frac{276}{100} = 2.76$ $1.75 \cdot 2.76 = 4.83$ \$4.83 per package
Cylinder 	\$2.25 per 100 square inches	$\frac{301.6}{100} = 3.016$ $2.25 \cdot 3.016 = 6.786$ \$6.79 per package
Cube 	\$1.25 per 100 square inches	$\frac{864}{100} = 8.64$ $1.25 \cdot 8.64 = 10.8$ \$10.80 per package



When students are completing question 3, monitor that students are dividing the surface area by 100 to start calculating the cost of the packaging.

4. If Manuel wants to use the cheapest packaging possible for his client, which packaging should he choose?
Manuel should choose the rectangular prism shaped package. It costs \$4.83 per package and the cheapest packaging.

9.9.3 Guided Instruction: Real-Life Applications of Surface Area

Component Overview: 9.9.3 Guided Instruction provides a scenario where students are determining how much frosting to use on two types of cakes. In questions 1-3 students compare the amount of frosting needed for cakes that are different figures and composite figures.



9.9.3 Guided Instruction: Real-Life Applications of Surface Area

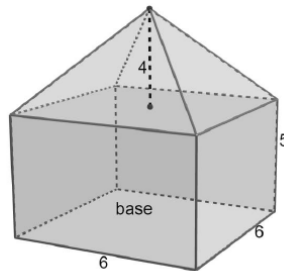
Jeni owns a bakery and is making two cakes.

Cake A	Cake B
Cake A is in the shape of a rectangular prism that is 3 inches thick, 9 inches long, and 12 inches wide.	The cake is in the shape of a cylinder and has a diameter of 8 in. The cake is composed of two layers, each 3 in. tall.

- When Jeni frosts the cake, she applies the frosting to all surfaces except for the bottom of the cake. She also frosts between each layer. Determine the amount of frosting Jeni needs to apply to each cake. Show your work.

Cake A	Cake B
$SA = B + Ph$ $SA = (12 \cdot 9) + (42 \cdot 3)$ $SA = 108 + 126$ $SA = 234 \text{ in}^2$	$SA = 2(B + Ch)$ $SA = 2(\pi(4)^2 + 2\pi(4)(6))$ $SA = 2(40\pi)$ $SA = 251.327 \text{ in}^2$

- Jeni makes a two-layer cake, as shown, where the measurements are in inches (in.) She will frost the whole cake except for the base of the bottom layer. She also adds frosting between layers.
 - Determine the total surface area that Jeni needs to frost.



The cake is a composite figure of a pyramid and a rectangular prism.

$$SA \text{ of Pyramid} + LA \text{ of Rect. Prism} = \text{Total SA}$$

$$SA = B + \frac{1}{2}Ph_s$$

$$h_s = 5$$

$$SA = 36 + \frac{1}{2}(24)(5)$$

$$SA = 96 \text{ in}^2$$

$$LA = Ph$$

$$LA = (24)(5)$$

$$LA = 120 \text{ in}^2$$

$$\text{Total SA} = 96 + 120 = 216 \text{ in}^2$$



Consider assigning students either Cake A or Cake B. After students have worked their question, pair each student in Group A with a student from Group B to share their results with each other.



Before students work on question 1, ask the following to prompt student thinking and responses:

- If the cakes are not frosted on the bottom, how does this change the formula to find the surface area?

9.9.3 Guided Instruction: Real-Life Applications of Surface Area (continued)

- b. Jeni's recipe for buttercream icing typically frosts a round cake with an 8 in. diameter that is 1.5 in. tall. How many batches of frosting will Jeni need to frost the two-layer cake in part A?

$$SA \text{ (of one batch)} = (B + Ch)$$

$$= \pi(4)^2 + 2\pi(4)(1.5)$$

$$= 87.965 \text{ in}^2$$

$$\text{Number of batches} = \frac{SA \text{ of cake}}{SA \text{ of one batch}} = \frac{216}{87.965} = 2.45$$

batches

Jeni will need 3 batches of frosting to frost the two layer cake.



Common Misconception

Students may include the base of the prism when finding the surface area that will be iced with frosting. Monitor students' work and ask probing questions about whether to include the base.



What part of the cake is frosted when Jeni frosts the first layer?

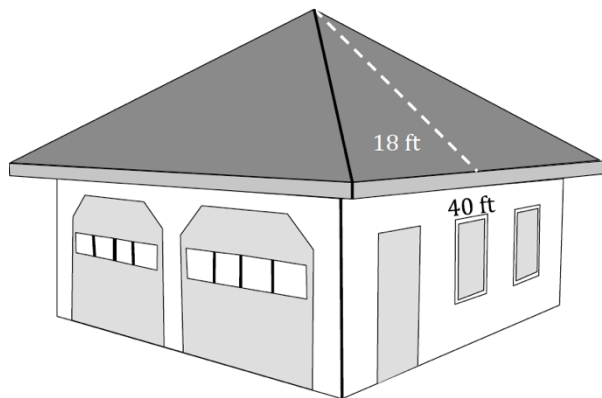
9.9.4 Guided Practice: Real-Life Applications of Surface Area

Component Overview: Students work through two real-life application questions involving the roofs of a garage and an art gallery.



9.9.4 Guided Practice: Real-Life Applications of Surface Area

- Lourdes' garage is in the shape of a square pyramid. The garage is shown, with dimensions given in feet (ft).



Lourdes is replacing the roof and installing shingles. In roofing terminology, a "square" is 100 square feet. Composition shingles come in bundles that cover $\frac{1}{3}$ of a square.

How many bundles should Lourdes buy to cover the roof?

Excluding the area of the base of the pyramid, use the lateral area formula, $LA = \frac{1}{2}Ph_s$.

$$LA = \frac{1}{2}(160)(18) = 1,440 \text{ ft}^2$$

They will need $\frac{1,440}{100} = 14.4$ "squares"

They will need $\frac{14.4}{(\frac{1}{3})} = 43.2$, which is about 44 bundles of shingles.



If students determine they need 4,320 bundles, then they failed to divide the lateral area by 100 to find the number of squares in the lateral area.

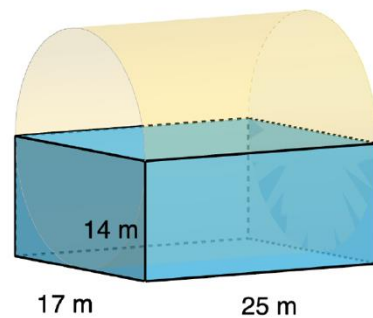
If students determine they need 4.8 or 5 bundles, then they multiplied by $\frac{1}{3}$ instead of dividing by $\frac{1}{3}$ (or multiplying by 3).



Why is lateral area calculated instead of surface area to find the number of shingles? (The roofing is only for the part of the roof outside of the house, which is lateral area.)

9.9.4 Guided Practice: Real-Life Applications of Surface Area (continued)

2. An architect is designing an art gallery. She used computer software to create a rough sketch of her idea. Her sketch is shown with measurements in meters (m). She would like for the art gallery to be all glass. To determine the estimated cost, she would need to find the area of all of the surfaces except for the floor. Determine the total surface area, in square meters, of the glass in the art gallery. Round to the nearest thousandth.



This is a composite figure of $\frac{1}{2}$ of a cylinder and the lateral area of a rectangular prism. Assume that the height of the rectangular prism is 14 meters.

$$\frac{1}{2} \text{ SA of Cylinder} + \text{LA of Rect. Prism} = \text{Total SA}$$

$$SA = \frac{1}{2}(2B + Ch)$$

$$SA = \pi r^2 + \pi rh$$

$$SA = \pi(8.5)^2 + \pi(8.5)(25)$$

$$SA \approx 284.75\pi \text{ m}^2$$

$$LA = Ph$$

$$LA = (84)(14)$$

$$LA = 1,176 \text{ m}^2$$

$$\text{Total SA} = 2,070.569 \text{ m}^2$$



As students are working through question 2, consider asking students the following question to prompt thinking:

- Why does the surface area of the cylinder need to be multiplied by $\frac{1}{2}$?



For students who finish early, challenge them to create a composite shape and scenario and then solve the question. If extra time allows, provide this question to other students who finish early.

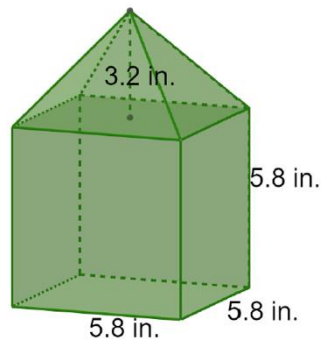
9.9.5 Wrap-Up: Ticket Out the Door

Component Overview: Students complete a *Ticket Out the Door* to demonstrate mastery on this lesson's learning target. Students should be allowed to access a calculator to complete 9.9.5 Wrap-Up.



9.9.5 Wrap-Up: Ticket Out the Door

- Jillian was building a birdhouse. Some of the dimensions, in inches (in.), of the birdhouse are shown. Determine the total amount of wood, in square inches, that Jillian will need to build the outside of the birdhouse.



$$\text{LA of Pyramid} + \text{SA of Rect. Prism with 1 base} = \text{Total SA}$$

$$LA = \frac{1}{2}Ph_s$$

$$LA = \frac{1}{2}(23.2)(\sqrt{18.65})$$

$$LA = 50.095$$

$$SA = B + Ph$$

$$SA = 33.64 + 134.56$$

$$SA = 168.2$$

$$SA = 218.295 \text{ in}^2$$



Students only need to find the lateral area of the pyramid and only use one base when finding the surface area of the rectangular prism.